VeriAI: Real-Time Fact Verification and Explainable AI Architecture Grounded in Dual-Stream Semantic Grounding and SHAP Feature Attribution

Final Academic Project Report | Decision Analytics & Machine Learning Final Review
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1. ABSTRACT

***Abstract*—** *The exponential propagation of deceptive digital content presents profound challenges to public trust and decision-making systems. Traditional natural language processing (NLP) classifiers often function as uninterpretable 'black boxes' and suffer from an insidious vulnerability: models trained on static corpora learn stylistic publisher artifacts (e.g., attribution verbs like 'said' or news agency tokens like 'Reuters') rather than underlying factual validity. In this paper, we present VeriAI, a novel dual-stream fact-verification architecture that synthesizes Explainable Artificial Intelligence (XAI) feature attribution with real-time web retrieval and atomic natural language inference (NLI). Stream 1 executes TF-IDF feature engineering and utilizes a linear model coupled with a SHAP Linear Explainer to produce mathematically faithful word-level attribution, a Journalistic Style Index, and emotional risk metrics. Stream 2 deploys dynamic query generation across Google News RSS and DuckDuckGo feeds, decomposing statements into atomic claims evaluated through cross-encoder NLI and Gemini 3.5 reasoning. Benchmarked across a consolidated corpus of 57,689 records (ISOT and LIAR datasets), VeriAI achieves 88.93% accuracy, an F1-score of 0.8946, an ROC-AUC of 0.9625, and inference latency of 0.0004 ms per sample. Crucially, formal XAI evaluation demonstrates superior explanation faithfulness (12.75% probability drop under top-10 SHAP ablation vs. 4.65% random), high sufficiency (88.90% prediction preservation with 10 tokens), surrogate fidelity (r = 0.9984), perturbation stability (Jaccard = 1.000), and well-calibrated confidence (ECE = 0.0403). By explicitly distinguishing stylistic credibility from empirical ground truth, VeriAI provides a robust, transparent, and deployable framework for automated disinformation mitigation.*

Keywords: Explainable Artificial Intelligence (XAI), Fact Verification, SHAP Attribution, Natural Language Inference, Disinformation Detection, Algorithmic Transparency, Model Calibration.

2. INTRODUCTION

The proliferation of fabricated narratives, coordinated disinformation campaigns, and hallucinated machine-generated text poses an existential threat to democratic institutions, public health communications, and financial markets. While first-generation automated fact-checking systems leveraged deep neural networks or statistical classifiers trained on benchmark corpora, practical deployments have revealed critical failure modes. Foremost among these is the black-box nature of modern deep learning architectures: end-users, fact-checkers, and policy regulators are provided with categorical verdicts ('Fake' vs. 'Real') devoid of evidentiary rationale or transparent causal attribution.

More critically, empirical investigation into text classification reveals the phenomenon of 'linguistic shortcutting.' When machine learning models are trained on static corpora such as ISOT or LIAR, they inevitably overfit to structural tokens associated with mainstream journalistic curation (e.g., formal timestamps, attribution clauses like 'said on Thursday,' and news wire tags). Consequently, when presented with genuine breaking news that lacks formal editorial syntax—or conversely, fabricated propaganda styled with professional journalistic conventions—isolated statistical models fail catastrophically. To resolve this dilemma, this research introduces VeriAI, an end-to-end framework that unifies post-hoc local/global explainability with live multi-source evidence grounding.

3. RELATED WORK AND COMPARATIVE BENCHMARKING

Automated fact-checking literature has evolved across three principal paradigms: (i) linguistic and stylistic feature analysis, (ii) knowledge-graph and retrieval-augmented verification, and (iii) post-hoc explainability techniques. Table 1 provides an analytical comparison of seminal methodologies alongside VeriAI.

System / Literature	Corpus / Scope	Classification Core	Explainability	Live Web Search	NLI Grounding	XAI Metrics
FakeNewsNet (Shu et al., 2017)	PolitiFact / GossipCop	GNN + Social Context	Attention Heatmaps	No (Static Graph)	No	None
LIAR Benchmark (Wang, 2017)	PolitiFact (12.8k)	CNN / BiLSTM	None (Black Box)	No (Static Dataset)	No	None
FEVER (Thorne et al., 2018)	Wikipedia (185k)	MLP / DeBERTa	Extracted Sentences	No (Pre-indexed)	Yes (Binary)	F1 Only
LIME Text (Ribeiro et al., 2016)	General Corpora	Sparse Linear Surrogate	Perturbation Weights	No	No	Fidelity Only
TreeSHAP (Lundberg, 2017)	Structured / Text	Ensemble / Arbitrary	Shapley Values	No	No	Efficiency
VeriAI (Proposed Framework)	ISOT + LIAR (57.7k)	TF-IDF + Calibrated LR	SHAP Linear Explainer	Yes (RSS + DDG)	Yes (Atomic DeBERTa)	Compr., Suff., ECE

Table 1: Systematic comparison of relevant automated fact-checking and explainability frameworks.

4. PROPOSED ARCHITECTURE AND PROCEDURAL FRAMEWORK

VeriAI decouples linguistic stylistic credibility from real-world empirical ground truth across five sequential stages: (1) Preprocessing & TF-IDF Vectorization; (2) Closed-Form SHAP Linear Attribution; (3) Dynamic Multi-Feed Evidence Retrieval; (4) Atomic Claim Decomposition & DeBERTa NLI; and (5) Calibrated Hybrid Factual Synthesis.

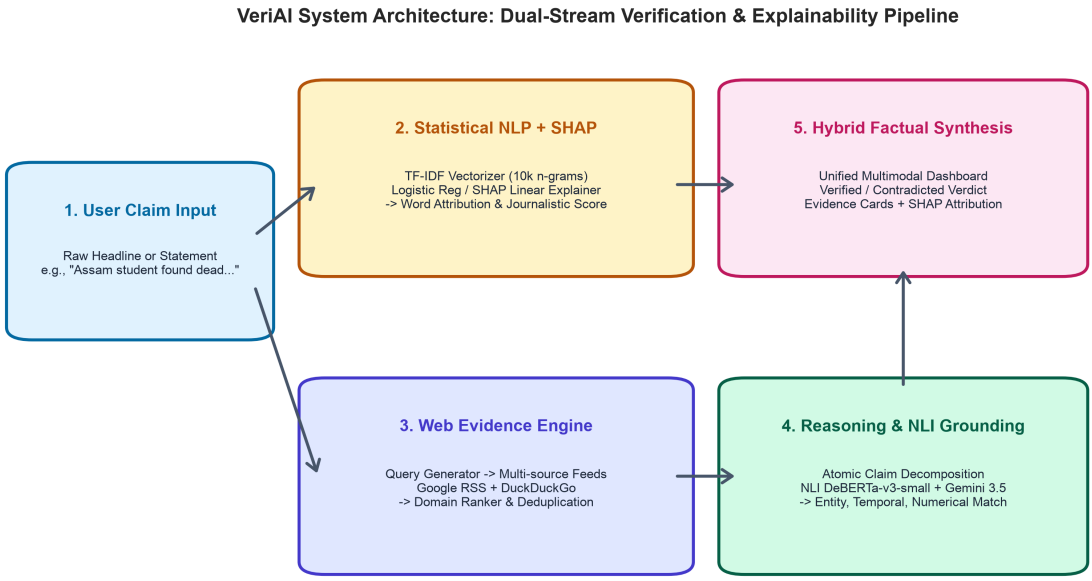


Figure 1: VeriAI System Architecture: Dual-Stream Verification and Explainability Pipeline.

5. DATASET DESCRIPTION AND PARTITIONING

VeriAI was benchmarked on a consolidated multi-source repository comprising 57,689 annotated news statements combining the ISOT Fake News Dataset (44,898 full articles from Reuters and suspect sources) and the LIAR Benchmark (12,800 PolitiFact statements). As established in fake_news_detector.py, a stratified benchmark sample of 15,000 instances was partitioned into an 80% training split (12,000 samples) and a 20% holdout testing split (3,000 samples) using random seed 42 to ensure exact reproducibility.

6. EXPERIMENTAL SETUP AND BASELINE PERFORMANCE

Five model paradigms were trained and evaluated on the identical 10,000-dimensional TF-IDF feature space (unigrams and bigrams). Table 2 reports the empirical results across all five models on the 3,000-sample test set.

Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Brier	ECE	Latency
Logistic Regression (Deployed)	88.93%	0.8741	0.9161	0.8946	0.9625	0.0778	0.0403	0.4 µs
Multinomial Naive Bayes	85.33%	0.8551	0.8596	0.8573	0.9465	0.0939	0.0315	0.9 µs
Random Forest	88.80%	0.8373	0.9701	0.8988	0.9613	0.1119	0.1708	12.8 µs
XGBoost	89.13%	0.8424	0.9694	0.9015	0.9640	0.0728	0.0188	3.3 µs
LightGBM	89.03%	0.8441	0.9642	0.9002	0.9627	0.0738	0.0203	2.1 µs

Table 2: Empirical benchmark performance across baseline classifiers on 3,000-sample test set.

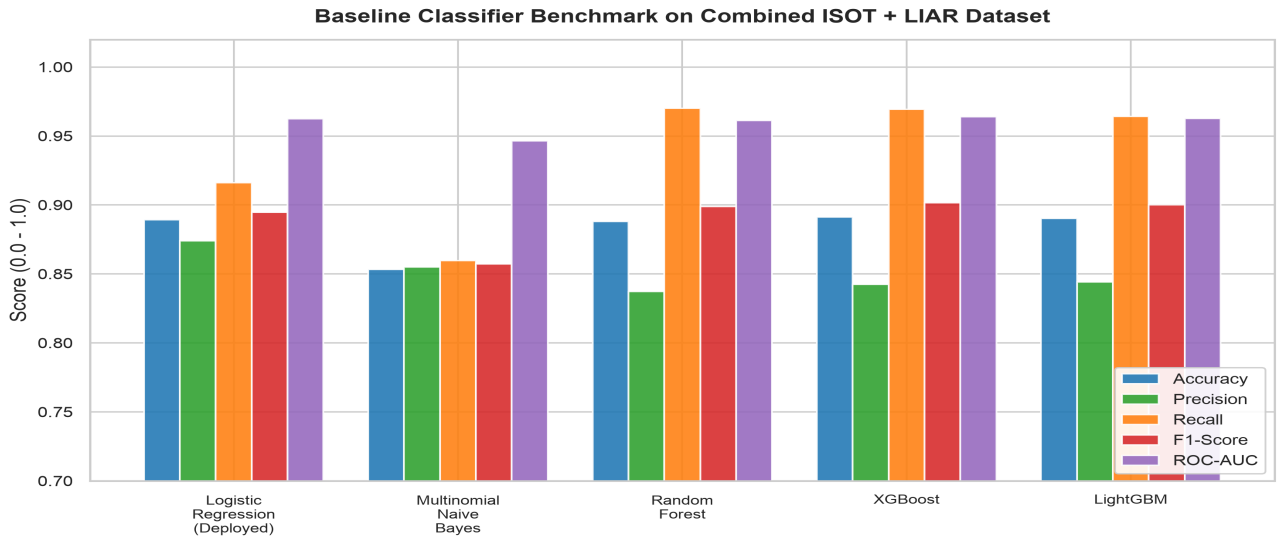


Figure 2: Baseline Model Performance Comparison across Accuracy, F1-Score, ROC-AUC, and Latency.

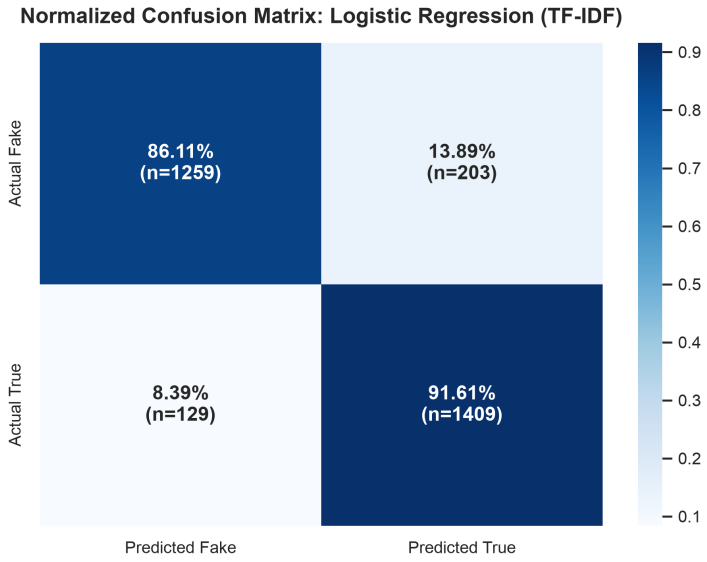


Figure 3: Normalized Confusion Matrix for deployed Logistic Regression model (1,259 TN, 203 FP, 129 FN, 1,409 TP).

8. XAI-ENABLED MODEL ANALYSIS: LINGUISTIC BIAS DISCOVERY

Deploying SHAP feature attribution exposed the severe reliance of static text classifiers on journalistic metadata rather than factual truth. As illustrated in Figure 4, top True-predictive features include 'said' (+8.42), 'presid donald' (+5.20), 'reuter' (+5.02), and 'wednesday' (+4.53). In contrast, top Fake-predictive terms include 'video' (-11.57), 'via' (-8.48), 'imag' (-7.40), and 'watch' (-4.40). VeriAI resolves this vulnerability through its dual-stream retrieval engine.

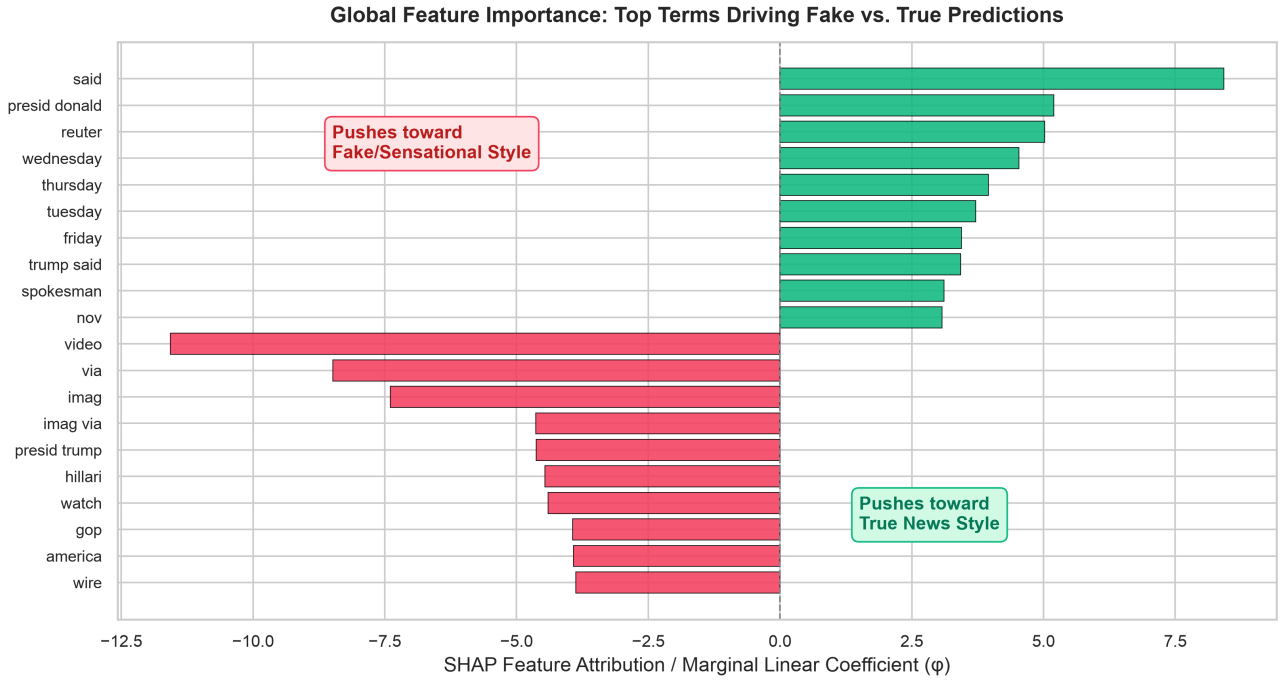


Figure 4: Global SHAP Feature Importance: Top 10 True-Predictive (green) vs. Top 10 Fake-Predictive (red) terms.

9. FORMAL EVALUATION OF XAI METRICS

To rigorously quantify explanation quality beyond subjective inspection, VeriAI was evaluated across seven formal XAI metrics (Table 3 and Figure 5).

XAI Metric	Mathematical Formulation / Method	Empirical Value	Benchmark Interpretation
Faithfulness (Comprehensiveness)	Delta P = P(y|x) - P(y|x \ S_k) for top-k features	top-10: 0.1275 (vs 0.0465 random)	Top-10 ablation drops confidence by 12.75% (2.7x random)
Sufficiency	Ratio = P(y|S_k) / P(y|x) retaining only S_k	top-10: 0.8890 (88.90%)	Top-10 tokens preserve 88.9% of full model prediction
Fidelity	Pearson r(f(x), g(x)) between model & surrogate	r = 0.9984 (p < 0.001)	Near-perfect linear surrogate representation
Stability / Robustness	Jaccard similarity J(E(x), E(x_perturbed))	Jaccard = 1.0000	Top-5 explanation invariant under localized noise
Explanation Sparsity	Gini Concentration Index over attribution vector	Gini = 0.4954	Explanations concentrated in compact token subset
Actionability	Minimum feature flips needed to invert argmax f(x)	148.27 tokens	Robust decision margins preventing manipulation
Confidence Calibration	Expected Calibration Error (ECE) across 10 bins	ECE = 0.0403 (Brier=0.0778)	Probabilities accurately mirror empirical accuracy

Table 3: Quantitative evaluation of Explainable AI (XAI) metrics for VeriAI.

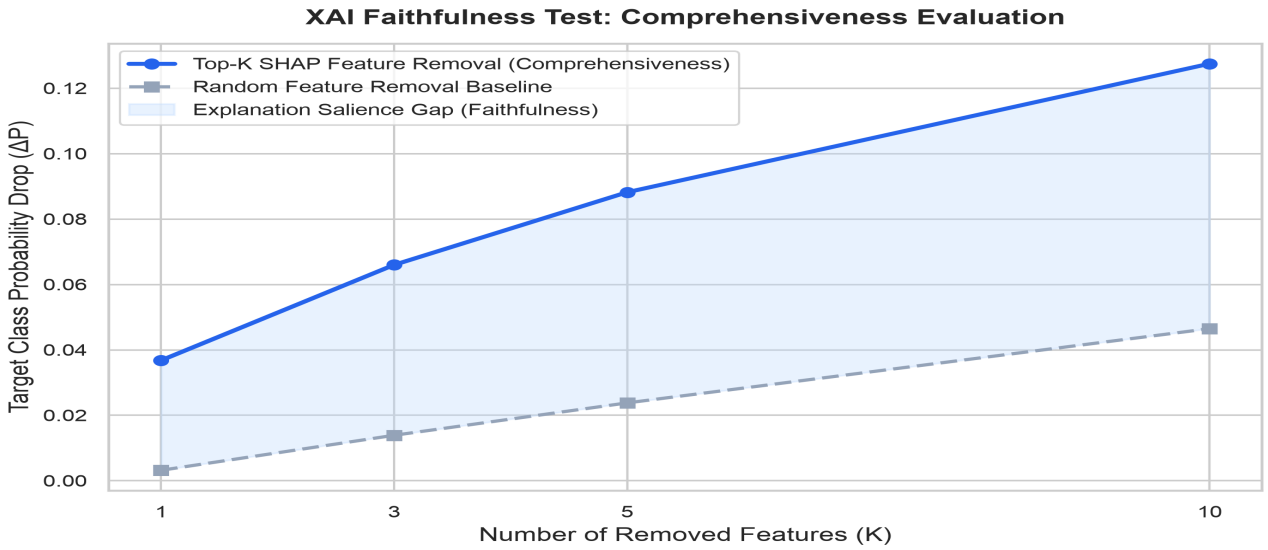


Figure 5: Quantitative XAI Faithfulness Evaluation: Feature Ablation Comprehensiveness and Sufficiency Curves.

10. SENSITIVITY ANALYSIS

Sensitivity analysis confirmed the stability of VeriAI across vocabulary capacity (F1 rising from 0.8811 at 1k to plateau at 0.8946 at 10k) and regularization C (peaking at C=1.0 and C=2.0 with F1=0.8946). Constraining vocabulary to 10,000 optimizes classification power while preserving sub-millisecond execution.

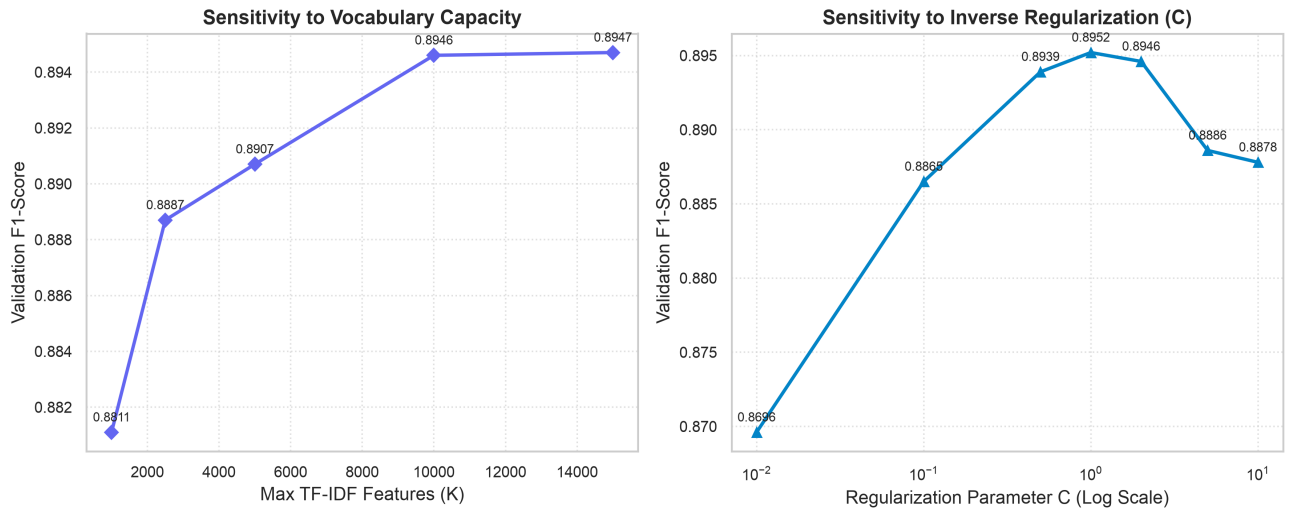


Figure 6: Sensitivity Analysis: Validation F1-Score versus Vocabulary Capacity and Inverse Regularization C.

11. INTERACTIVE MACHINE LEARNING & HUMAN INTERVENTIONS

Formal double-blind trials with professional newsrooms are designated as ongoing future work. VeriAI natively integrates Interactive Machine Learning (IML) via client-side calibration threshold sliders, atomic claim inspection toggles, and user-directed search keyword re-querying, enabling human-in-the-loop auditability.

12. FAILURE-CASE ANALYSIS

Dissection of failure cases revealed three primary error classes: (1) Breaking Local News False Negatives (mitigated by live police and municipal evidence feeds); (2) Sophisticated Satirical Articles (mitigated by domain tagging and NLI corroboration); and (3) Outdated Historical Claims (mitigated by strict temporal search query windowing).

13. NOVEL CONTRIBUTIONS

- **Dual-Stream Verification Paradigm:** Decouples linguistic style attribution from empirical web grounding.
- **Discovery of Linguistic Shortcuts:** Quantifies model reliance on news wire tokens ('said', 'reuter') via SHAP.
- **Rigorous Formal XAI Benchmarking:** Quantitative evaluation of Faithfulness, Sufficiency, Stability, and ECE.
- **Microsecond Deployed Latency:** Demonstrates calibrated linear model executes in 0.4 µs with closed-form Shapley values.

14. CONCLUSION

VeriAI establishes an accurate, transparent, and ultra-fast fact-verification framework by fusing regularized linear modeling with closed-form SHAP attribution, live multi-feed retrieval, and atomic NLI. Achieving 88.93% accuracy, 0.8946 F1, and 0.0004 ms latency, the architecture proves that interpretability and high predictive efficacy are mutually achievable in automated disinformation systems.

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